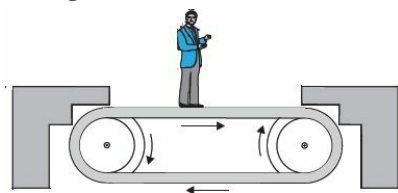
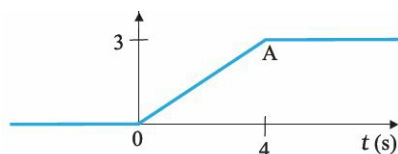


CBSE Test Paper 02
Chapter 5 Laws of Motion

1. Figure shows a man standing stationary with respect to a horizontal conveyor belt that is accelerating with 1 m s^{-2} . What is the net force on the man? Mass of the man = 65 kg **1**



- a. 70 N
b. 55 N
c. 65 N
d. 60 N
2. Figure shows the position-time graph of a particle of mass 4 kg. What is the impulse at $t = 0$ and $t = 4 \text{ s}$? (Consider one-dimensional motion only) **1**

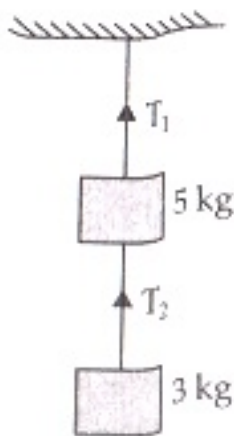


- a. 3 kg m s^{-1} at $t = 0$ and -3 kg m s^{-1} at $t = 4 \text{ s}$
b. 2 kg m s^{-1} at $t = 0$ and -2 kg m s^{-1} at $t = 4 \text{ s}$
c. 5 kg m s^{-1} at $t = 0$ and -3 kg m s^{-1} at $t = 4 \text{ s}$
d. 4 kg m s^{-1} at $t = 0$ and -4 kg m s^{-1} at $t = 4 \text{ s}$
3. A body of mass 5 kg is acted upon by two perpendicular forces 8 N and 6 N. Give the magnitude and direction of the net force acting on the body. **1**
- a. 16 N at an angle of 46° to the 8 N force
b. 10 N at an angle of $36^\circ 52'$ towards the 8 N force
c. 14 N at an angle of 37° to the 8 N force
d. 12 N at an angle of 38° to the 8 N force
4. In which of the following cases is the net force non zero? **1**
-

-
- a. a drop of rain falling down with a constant speed
 - b. a cork of mass 10 g floating on water
 - c. a coin falling under gravity
 - d. a kite skillfully held stationary in the sky
5. Give the magnitude and direction of the net force acting on a stone of mass 0.1 kg lying on the floor of a train which is accelerating with 1 m s^{-2} , the stone being at rest relative to the train. Neglect air resistance. **1**
- a. 1.0 N in the direction of motion
 - b. 0.1 N opposite to the direction of motion
 - c. 0.1 N along the direction of motion
 - d. 0.2 opposite to the direction of motion
6. A soda water bottle is falling freely. Will the bubbles of the gas rise in the water of the bottle? **1**
7. A body is acted upon by a number of external forces. Can it remain at rest? **1**
8. The two ends of spring - balance are pulled each by a force of 10 kg.wt. What will be the reading of the balance? **1**
9. A passenger of mass 72.2 kg is standing on a weighing scale in an elevator. What does the scale read when the elevator cab is **2**
1. descending with constant velocity?
 2. ascending with constant acceleration, 3.5 m/s^2 ?
10. Give the magnitude and direction of the net force acting on a stone of mass 0.1 kg, Neglect air resistance throughout. **2**
- a. just after it is dropped from the window of a stationary train,
 - b. just after it is dropped from the window of a train running at a constant velocity of 36 km/h.
11. A block of mass M is held against a rough vertical wall by pressing it with a finger. If the coefficient of friction between the block and the wall is μ and the acceleration due
-

to gravity is g , calculate the minimum force required to be applied by the finger to hold the block against the wall? **2**

12. The driver of a three-wheeler moving with a speed of 36 km/h sees a child standing in the middle of the road and brings his vehicle to rest by the brake in 4.0 s just in time to save the child. What is the average retarding force on the vehicle by the brake? The mass of the three-wheeler is 400 kg and the mass of the driver is 65 kg. **3**
13. Two masses of 5 kg and 3 kg are suspended with help of massless inextensible strings as shown in Figure. Calculate T_1 and T_2 when whole system is going upwards with acceleration $= 2 \text{ ms}^{-2}$ (use $g = 9.8 \text{ ms}^{-2}$) **3**



14. Ten one rupee coins are put on top of one another on a table. Each coin has a mass m kg. Give the magnitude and direction of **3**
- The force on the 7th coin (counted from the bottom) due to all coins above it.
 - The force on the 7th coin by the eighth coin (counted from the bottom) and
 - The reaction of the sixth coin on the seventh coin (counted from the bottom).
15. The rear side of a truck is open and a box of 40 kg mass is placed 5 m away from the open end. The coefficient of static friction between the box and the surface below it is 0.15. On a straight road, the truck starts from rest and accelerates with 2 ms^{-2} . At what distance from the starting point does the box fall off the truck? (Given, $g = 10 \text{ m/s}^2$) **5**

